Mass as a Time: Gravitational Time, Clock-Rate Depth, and the Schuler Scale

Light-Meter Physics

Abstract

The gravitational parameter GM can be expressed as a characteristic time $\tau_g \equiv GM/c^3$. This time is neither an elapsed proper time nor a clock period; it is a scale formed from mass, gravity, and the speed of light. For a spherical model of radius R, the radius light-time $t_R \equiv R/c$ gives the compactness $x_R = \tau_g/t_R$. In a static Schwarzschild exterior, the inverse clock-rate excess is exactly $(1-2x_R)^{-1/2}-1$ and equals x_R only to first order. In the Newtonian Schuler model, $\omega_{\mathrm{S}}^2 = GM/R^3 = \tau_g/t_R^3$. These definitions produce three related scales: τ_g/t_R is dimensionless, $\tau_g/t_R^2 = g_{\mathrm{N}}/c \equiv f_g$ is an inverse-time scale, and $\tau_g/t_R^3 = \omega_{\mathrm{S}}^2$. An idealized Earth calculation gives $\tau_g \simeq 14.79$ ps, $t_R \simeq 21.25$ ms, and a first-order surface-clock lag of about 60.1 μs per reference day. The relationships are dimensional and algebraic, not a causal link between navigation and gravity.

Keywords: gravitational time scale, clock rate, Schwarzschild spacetime, Schuler tuning, weak field

1 Introduction

The combination GM/c^3 has units of time and appears naturally when gravitational effects are written in relativistic units. Calling it "mass as a time" is convenient shorthand, but it does not mean that mass itself has become elapsed time or that a clock accumulates this interval. The quantity is a characteristic scale built from the gravitational parameter GM and the exact speed of light c.

This paper relates that scale to two others. The first is the light-time equivalent R/c of a chosen radius. The second is the inverse angular frequency of a Newtonian Schuler mode. Combining them recovers the weak-field static clock-rate comparison and the acceleration

scale g_{N}/c, here denoted f_g. The derivation keeps the exact Schwarzschild clock-rate formula distinct from its first-order approximation and keeps Schuler dynamics Newtonian.

The model uses an ideal, spherical, nonrotating source with gravitational parameter GM and radius R. For the exact static-clock expression, R is an areal radius in the vacuum Schwarzschild exterior and must lie outside the source. The reference time t is normalized to proper time at infinity; τ is the proper time of the static clock at R. Numerical Earth values are illustrative spherical-model inputs, not a geodetic clock-transfer model. Rotation, oblateness, altitude variation, other bodies, and the geoid are omitted.

2 The gravitational time scale

Define

$$\tau_g \equiv \frac{GM}{c^3}. \tag{1}$$

Since $[GM] = \mathrm{m}^3\,\mathrm{s}^{-2}$ and $[c^3] = \mathrm{m}^3\,\mathrm{s}^{-3}$, τ_g has units of seconds. The term "mass time" refers to this gravitational scale, not to a mass M converted to seconds without the factor G.

The Schwarzschild radius is

$$r_s \equiv \frac{2GM}{c^2} = 2c\tau_g. \tag{2}$$

Thus τ_g is half the Schwarzschild radius divided by c. This is a relation between scales; it does not imply that a body has that radius or that light physically traverses an interior path. The same combination occurs in relativistic pulsar timing, where the solar mass time $T_\odot = GM_\odot/c^3$ is used in timing parameters and gravitational propagation delays [6].

3 Radius light-time and the static clock

For a chosen areal radius R, define the radius light-time

$$t_R \equiv \frac{R}{c}. \tag{3}$$

This is a unit conversion from length to time, not a claim about a signal traveling through the body. Their ratio is the dimensionless compactness

$$x_R \equiv \frac{\tau_g}{t_R} = \frac{GM}{Rc^2}. \tag{4}$$

In the Schwarzschild exterior, a static clock at radius R has lapse $\alpha(R) = \mathrm{d}\tau/\mathrm{d}t = \sqrt{1 - r_s/R}$ [3]. Define its dimensionless inverse rate and excess by

$$\delta\tau(R) \equiv \frac{\mathrm{d}t}{\mathrm{d}\tau} = \frac{1}{\alpha(R)}, \qquad \epsilon_{\mathrm{grav}}(R) \equiv \delta\tau(R) - 1. \tag{5}$$

Here $\delta\tau$ is a rate ratio, not an elapsed time interval. With t normalized at infinity, the exact static relation becomes

$$\delta\tau(R) = \left(1 - \frac{2\tau_g}{t_R}\right)^{-1/2}, \qquad \epsilon_{\mathrm{grav}}(R) = (1 - 2x_R)^{-1/2} - 1. \tag{6}$$

This expression applies to a static clock in the exterior, where $R > r_s$. The horizon boundary corresponds to $t_R = 2\tau_g$; a static observer cannot be placed there.

For $x_R \ll 1$, expansion of Eq. (6) gives

$$\epsilon_{\mathrm{grav}}(R) = x_R + \frac{3}{2}x_R^2 + \frac{5}{2}x_R^3 + \cdots \simeq \frac{\tau_g}{t_R}. \tag{7}$$

Only the final equality is a first-order approximation. The excess depends on the reference time normalization; it is not a locally observed absolute clock deficit. In the Newtonian limit it is also $-\Phi(R)/c^2$ when $\Phi(\infty) = 0$.

4 The Schuler timescale and the three powers

In the idealized linear Schuler model, a small navigation-error mode q obeys

$$\ddot{q} + \omega_{\mathrm{S}}^2 q = 0, \qquad \omega_{\mathrm{S}}^2 = \frac{g_{\mathrm{N}}}{R}, \tag{8}$$

where g_{N} is the Newtonian surface-gravity magnitude of the spherical source. For a point-mass exterior, $g_{\mathrm{N}} = GM/R^2$, so

$$\omega_{\mathrm{S}}^2 = \frac{GM}{R^3}. \tag{9}$$

This is the standard Schuler angular-frequency scale [1, 2]. It has the same form as the Newtonian angular frequency squared of a circular orbit at radius R; the equality is a consequence of the model equations, not a claim that the navigation error mode is a physical orbit. Real navigation systems include sensor errors, damping, rotation, and other terms beyond this ideal oscillator.

Let

$$\tau_{\mathrm{S}} \equiv \frac{1}{\omega_{\mathrm{S}}}, \qquad f_{\mathrm{S}} \equiv \frac{\omega_{\mathrm{S}}}{2\pi}, \qquad T_{\mathrm{S}} \equiv \frac{2\pi}{\omega_{\mathrm{S}}}. \tag{10}$$

The quantity $\tau_{\rm S}$ is the one-radian phase timescale, not the full Schuler period $T_{\rm S}$; $f_{\rm S}$ is the cyclic frequency. Using $t_R = R/c$ and $\tau_g = GM/c^3$ in Eq. (9) yields

$$\omega_{\rm S}^2 = \frac{\tau_g}{t_R^3}, \qquad t_R^3 = \tau_g \tau_{\rm S}^2. \tag{11}$$

The second form says that, on a logarithmic timescale, t_R lies two-thirds of the way from τ_g to $\tau_{\rm S}$:

$$\ln\!\left(\frac{t_R}{\tau_g}\right) = \frac{2}{3}\ln\!\left(\frac{\tau_{\rm S}}{\tau_g}\right). \tag{12}$$

The same mass time divided by successive powers of the radius light-time gives

$$\frac{\tau_g}{t_R} = \frac{GM}{Rc^2} = x_R, \tag{13}$$

$$\frac{\tau_g}{t_R^2} = \frac{GM}{R^2 c} = \frac{g_{\rm N}}{c} = \omega_{\rm S}^2 t_R, \tag{14}$$

$$\frac{\tau_g}{t_R^3} = \frac{GM}{R^3} = \omega_{\rm S}^2. \tag{15}$$

For compactness of notation, define

$$f_g \equiv \frac{g_{\rm N}}{c}. \tag{16}$$

Despite the symbol f, f_g denotes an inverse-time scale, not a cyclic frequency or an oscillation. No factor of 2π is part of its definition. Equations (13)–(15) are exact algebraic identities within the stated Newtonian definitions. The identification of τ_g/t_R with the static inverse-rate excess is only first order, as shown in Eq. (7).

5 A consistent Earth and Sun illustration

For the series' ideal spherical-Earth reference set, take $GM_\oplus = 3.98600435507 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$ [4], choose the volumetric mean radius $R_\oplus = 6.371000 \times 10^6$ m [5], and use the exact SI value $c = 299\,792\,458\ \mathrm{m\,s^{-1}}$ [7]. This defines one spherical model; the radius and gravitational parameter are not a claim that the real Earth is spherical. Rounded results are in Table 1.

Table 1: Characteristic times and compactness for idealized spherical models. The solar inputs are IAU nominal conversion constants.

Model	GM ($\mathrm{m^3\,s^{-2}}$)	R (m)	τ_g	t_R	$x_R = \tau_g/t_R$
Earth	$3.98600435507 \times 10^{14}$	6.371000×10^6	14.7937 ps	21.2514 ms	0.69613 ppb
Sun	1.3271244×10^{20}	6.957×10^8	4.92549 μs	2.32061 s	2.12250 ppm

The solar row uses the IAU nominal solar mass parameter and nominal radius [8]. These are defined conversion constants, not claims about the Sun's instantaneous measured mass or radius. For both rows, x_R equals the inverse-rate excess only in the weak-field approximation. The exact expression for an ideal Schwarzschild exterior remains Eq. (6).

For the selected Earth model,

$$g_{\mathrm{N}} = \frac{GM_\oplus}{R_\oplus^2} \simeq 9.82025\ \mathrm{m\,s^{-2}}, \qquad f_g = \frac{g_{\mathrm{N}}}{c} \simeq 3.27568 \times 10^{-8}\ \mathrm{s^{-1}} = 32.7568\ \mathrm{nHz}. \tag{17}$$

The Schuler quantities for the same model are

$$\omega_{\mathrm{S}} \simeq 1.24153 \times 10^{-3}\ \mathrm{s^{-1}}, \qquad \tau_{\mathrm{S}} \simeq 805.46\ \mathrm{s}, \qquad f_{\mathrm{S}} \simeq 197.60\ \mu\mathrm{Hz}, \qquad T_{\mathrm{S}} \simeq 84.35\ \mathrm{min}. \tag{18}$$

These use $g_{\mathrm{N}} = GM/R^2$ throughout. Substituting the conventional standard acceleration $g_{\mathrm{n}} = 9.80665\ \mathrm{m\,s^{-2}}$ instead would give $g_{\mathrm{n}}/c \simeq 32.710$ nHz, but that is a different input and not the surface field of this ideal spherical mass model [9].

6 What the mass time means for an interval

For a static clock in the weak field, $\mathrm{d}\tau/\mathrm{d}t \simeq 1 - x_R$. Over a reference-time interval Δt, its accumulated lag relative to that reference is therefore

$$\Delta t - \Delta\tau \simeq x_R\,\Delta t = \frac{\tau_g}{t_R}\,\Delta t. \tag{19}$$

If the reference interval is exactly one radius light-time, $\Delta t = t_R$, then

$$\Delta t - \Delta\tau \simeq \tau_g. \tag{20}$$

This equality is a first-order clock comparison, not an assertion that the mass time is stored in the clock or represents a separate physical delay.

For the Earth example, the first-order lag is about 60.15 μs per 86 400 s of reference time. Over one Schuler one-radian timescale it is about 0.561 μs; over one full Schuler cycle it is about 3.523 μs. Over one radius light-time, 21.2514 ms, it is approximately 14.7937 ps, equal to τ_g at this order. These are elapsed-interval comparisons between a static surface clock and the reference time, not operational clock errors.

7 The inverse-time scale and the static gradient

The relation to a spatial gradient can be shown directly in Schwarzschild coordinates [3]. In the radial direction, proper distance satisfies $\mathrm{d}L = \mathrm{d}r/\alpha$, so for a radial scalar $F(r)$,

$$\nabla F = \alpha \frac{\mathrm{d}F}{\mathrm{d}r}\,\hat{\mathbf{r}}, \tag{21}$$

where $\hat{\mathbf{r}}$ points outward. Applying this to the exact inverse rate gives

$$c^2 \nabla \ln \delta\tau = -\frac{GM}{r^2 \alpha(r)}\,\hat{\mathbf{r}}. \tag{22}$$

The vector points inward and is the negative of the proper acceleration required to hold a static position. In the weak field, $\alpha \simeq 1$, so its magnitude divided by c is $g_{\mathrm{N}}/c = f_g$ to leading order. Thus the middle scale in Eq. (14) is consistent with the static inverse-rate gradient, while the exact hovering acceleration includes the factor $1/\alpha$.

This gradient relation concerns a family of static observers and uses proper spatial distances. It does not turn f_g into an independently detectable oscillation frequency, nor does it make the clock-rate ratio a new field or material density. The connection in this paper is between standard relativistic clock comparisons and a Newtonian navigation timescale through shared model parameters and unit conversions.

8 Conclusion

The characteristic gravitational time $\tau_g = GM/c^3$ expresses the gravitational parameter as a time scale. Paired with the radius light-time $t_R = R/c$, it gives the compactness τ_g/t_R and, in the weak field, the static inverse-rate excess. In a Newtonian spherical-Earth Schuler model, the additional relation $\omega_{\mathrm{S}}^2 = GM/R^3$ yields $\tau_g/t_R^2 = g_{\mathrm{N}}/c \equiv f_g$ and $\tau_g/t_R^3 = \omega_{\mathrm{S}}^2$. The symbols organize dimensional relationships; they do not imply a causal mechanism between navigation and gravity. The exact static clock-rate formula is retained separately from its first-order approximation, and the inverse-time scale f_g is not an oscillation frequency.

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